

DEVICE ANATOMY 2.0

Engineering Investigation of an Electronic Device at End-of-Life

Course	E-Waste & Environmental Management	Class	TE INFT
Duration	45-60 minutes	Marks	10

10/10

Group Details

Group No.:	10	Device Investigated:	Hard disk
Date:	02-09-26	Condition:	Not working
Member 1:	Varun Singh	Member 2:	Rohan Singh
Member 3:	Vedant Chauhan	Member 4:	Aryan Acharya
Member 5:	Mrunal Wanjare		

Condition: ☐ Functional ☒ Non-functional ☐ Damaged ☐ Unknown

Activity Evidence

During the activity, photograph all group members actively participating in the Device Anatomy investigation. Print the photograph and attach it below. A posed group selfie is not considered activity evidence.

ATTACH PRINTED GROUP PHOTOGRAPH HERE

Photo caption: Our group investigating the anatomy and end-of-life characteristics of a hard disk.

Group members visible: Aryan, Rohan, Varun, Mrunal, Vedant

ACTIVITY CHALLENGE

You are an e-waste engineer. Your task is to investigate the material anatomy of an electronic product, identify recovery opportunities and environmental risks, and propose engineering changes that improve its end-of-life management.

Core question: Can you make this device more circular?

Part A — Device Profile

Device name	Hard disk
Primary function	Storing data
Why does this device become e-waste?	☐ Obsolescence ☒ Damage ☐ Upgrade ☐ Repair not economical ☒ Failure ☐ Other: _____

Part B — Engineering Anatomy

Identify at least 6 major components. Do not merely name the component — explain the material choice.

#	Component	Function	Material(s)	Why is this material suitable?
1	Disk	rotation for data store	Aluminium alloy	Lightweight with smooth surface.
2	Reader/ write head	Reads & writes data.	Silicon	Precise magnetic sensing
3	magnet	used to keep it track.	Neodymium alloy	very strong force in a small size
4	Platter	store data magnetically	Alloy with magnetic coats	Data storage because of coating
5	Spindle motor	rotation of the disk	Copper	Copper provide good conductivity.
6	PCB board	Controls motor	Copper Epoxy	Fiberglass / epoxy give good mechanical support. Copper conducts good electricity.

Engineering Challenge: Which component demonstrates the most interesting material selection? Why?

Permanent magnet - Neodymium is interesting because it produces a very strong field despite being very small.

Part C — Material Detective

Tick the major material categories present in your device:

- ☒ Ferrous metals ☐ Aluminium ☒ Copper ☐ Precious metals
☐ Engineering plastics ☒ Glass ☒ Ceramics ☐ Semiconductor materials
☒ Rubber/elastomers ☐ Other: _____

	Material 1	Material 2
Name	Aluminium alloy	Copper
Where is it used?	HDD platter	PCB tracks
Why is it used?	Lightweight	Electrical conductivity

	Material 1	Material 2
Can it be recovered/recycled? How?	Yes, by dismantling the HDD	Copper can be recovered during e-waste

Part D — Hidden Value: The Urban Mine

Identify the three most valuable/recoverable materials in your device.

Material	Component containing it	Potential value	Recovery difficulty
1. Neodymium	Permanent magnet	Low / Medium / High	Low / Medium / High
2. Copper	PCB & motor	Low / Medium / High	Low / Medium / High
3. Aluminium	Platters	Low / Medium / High	Low / Medium / High

Critical Thinking: Is the most valuable material necessarily the easiest to recover? Explain.

No neodymium has high values but recovering it can be difficult bcz magnet is integrated.

Part E — Hazard Detective

Identify components that may require controlled handling.

Component	Possible hazard	Risk if improperly disposed	Handling requirement
PCB	electronic chemicals	soil & water contamination	send to authorize e-waste recycler
Magnet	strong magnetic force	injury or dmg to sensitive device	handle carefully during dismantling
platter	sharp edges	cuts & possible environment	use gloves & controlled dismantling
electronic components	potential hazardous substances	potential hazardous substance pollution	process through certified e-waste facility

Most environmentally concerning component: PCB | controller board

Why?
The PCB contains many electronic materials and may contain substances that can contaminate soil and water if HDD is improperly disposed of. ∴ it should be sent to an authorized e-waste recycling facility rather than landfill.

Part F — End-of-Life Decision

Recommend the most appropriate pathway for each major component.

Component	Reuse	Repair	Refurbish	Recycle	Controlled Disposal
Platters	☐	☐	☐	☒	☐
Read/write Head	☐	☒	☒	☐	☐
Permanent magnet	☐	☐	☐	☒	☐
PCB	☐	☒	☒	☒	☐

Justify ONE of your decisions:

The permanent magnets should be recovered and recycled because neodymium is a valuable rare-earth material. Recovering the magnets reduces the need for new mining & allows the material to be reused in other products.

Part G — Design for Disassembly

Imagine that you are the engineer who designed this product. Identify three design features that make recycling, repair or disassembly difficult.

#	Existing design problem
Problem 1	The HDD contains many small screws and tightly packed components, making dismantling time-consuming.
Problem 2	The actuator & magnet assembly can be difficult to separate safely.
Problem 3	Different materials such as metals, plastics & electronic components are assembled together making material separation more difficult.

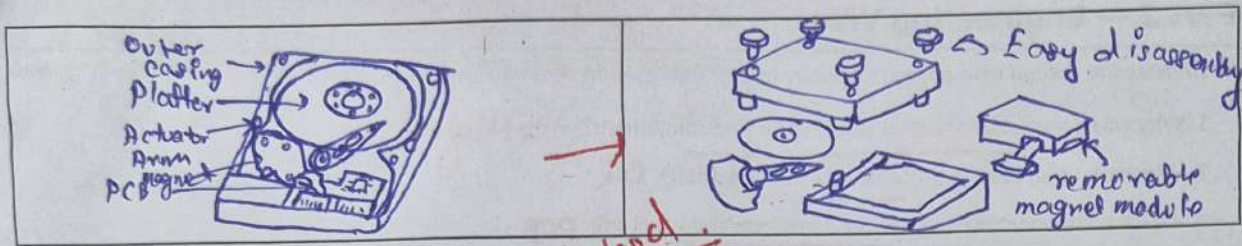
Now redesign it!

Propose three engineering modifications that would make the product more circular.

Existing Design Problem	Proposed Design Change	Sustainability Benefit
Many screws & difficult access	Use fewer standardized screws & clearly accessible fasteners	Faster repair and easier disassembly
Difficult magnet/actuator removal	Design the magnet assembly as a removable module	Easier recovery of valuable rare-earth magnets
Mixed material are difficult to separate	Use material labels & modular construction	Better sorting & recycling

Sketch your redesigned device/component — show at least one change that improves repair, disassembly or material recovery.

ORIGINAL	REDESIGNED
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Part H — E-Waste Journey

Trace the end-of-life journey of ONE component.

Component selected: Aluminium platters

DEVICE	CONSUMER DISCARD	COLLECTION	DISMANTLING	MATERIAL RECOVERY	NEW PRODUCT / RESOURCE
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What could go wrong at any stage?

If HDD is thrown into general waste, valuable materials may be lost & electronic components can contribute to environment pollution.

Part I — Circular Economy Decision

Our device should NOT simply be treated as waste because...

An HDD contains valuable & recoverable materials such as Al, Cu, & rare earth magnets. A working HDD can also potentially be reused, while a failed HDD can be dismantled & recycled to recover its materials.

Which circular strategy would have the greatest impact?

☒ Repair ☒ Reuse ☐ Refurbishment ☐ Remanufacturing

☒ Recycling ☒ Material recovery

Why?

Reuse has the greatest impact when the HDD is still functional because extending its useful life avoids the environmental impacts of manufacturing a replacement drive & keeps the existing materials in use for longer. Before reuse, the old data must be securely erased.

Part J — Engineering Viva

Be prepared to defend your answers. Your instructor may ask any of the following:

1. Why was a particular material selected for one component? → Magnet ~~N~~
2. Which material has the greatest recovery potential? Cu
3. Which component is most difficult to recycle and why? → PCB
4. What is the single most effective design modification you propose?
5. If your sustainable redesign increases manufacturing cost by 8%, would you still recommend it? Why?

Final Group Verdict

The most valuable component is:	Permanent magnet / Neodymium magnet
The component with the greatest environmental concern is:	PCB / Controller board
The biggest design-for-recycling problem is:	Difficult disassembly and separation of different materials and components
Our most important redesign recommendation is:	Use a modular HDD design with standardized screws and easily removable components
ONE-SENTENCE ENGINEERING VERDICT We recommend designing the HDD for easy disassembly because it extends product life and allows valuable materials such as copper, aluminium and rare earth magnets to be recovered instead of becoming e-waste. repair and material recovery	

Assessment Rubric — 10 Marks

Criterion	Excellent (2)	Good (1.5)	Developing (1)	Marks
1. Engineering Anatomy	Accurate component + function + material analysis	Mostly accurate	Basic identification	/2
2. Material & Recovery Analysis	Strong material/recovery reasoning	Adequate reasoning	Limited reasoning	/2
3. Hazard & E-waste Assessment	Identifies risks with clear justification	Identifies major risks	Basic identification	/2
4. Design for Disassembly	Innovative, feasible engineering solutions	Relevant solutions	Basic suggestions	/2
5. Engineering Defence/Viva	Clear, evidence-based justification	Reasonable explanation	Limited justification	/2
TOTAL				10

Reflection

What surprised you most about the materials hidden inside your device?

~~the~~ the ~~power~~ Neodymium magnet bcz of its power

After this activity, would you still call this device "waste"? Why?

Not completely

KEY TAKEAWAY

E-waste is not merely waste — it is a complex mixture of materials, components, resources and risks that requires engineering solutions for responsible recovery.

